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Magnetic fabric and paleomagnetism of the Middle Triassic siliciclastic rocks from the Nanpanjiang Basin, South China: Implications for sediment provenance and tectonic process

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ABSTRACT

A combined magnetic fabric and paleomagnetic study has been carried out on the siliciclastic rocks gathered from a stratigraphic cross-section through the Nanpanjiang Basin, South China, in an attempt to extract the paleoflow information preserved in and, thus, constrain the possible origins of these clastic rocks. The sediments used for this study were formed by sediment-gravity flows along the southern margin of the South China block in the Middle Triassic time (ca. 245–228 Ma). The results show a normal distribution of both low field magnetic susceptibility values and natural remanent magnetization intensities, which along with the monotonic detrital framework mode, mainly comprising quartz and lithic particles, may suggest a single provenance involved in deposition of these clastic deposits. Anisotropy of magnetic susceptibility (AMS) analysis acquires primarily the sedimentary magnetic fabrics, which, in this study, reveal paleoflow directions ranging from NNW to ENE with an overall mean orientation of NE. Demagnetization on a part of samples isolates a characteristic remanent component averaged at $D = 44.8°$, $I = 16.9°$, $\kappa = 9.7$, $\alpha_{95} = 6.5°$, $n = 55$, corresponding to a paleolatitude N8.6° and a clockwise rotation of ca. 45° since the Middle Triassic for the studied cross-section. This mean direction passes fold tests and is consistent with the reference direction expected from the South China block at the 95% confidence level. Restoring this ~45° declination renders an overall northward paleoflow, which, combined with other evidence, suggests a southern provenance for these sediments during deposition in the Middle Triassic time. In terms of the early Mesozoic plate framework of southeastern Asia, a tectonic scenario is proposed here, whereby the nearly N–S convergence of the Indochina and South China blocks and its related Indosinian orogeny in the Middle Triassic caused the formation of the Nanpanjiang foreland basin, which was filled by voluminous detritus shed from the uplifted orogenic belt on its southern side.

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1. Introduction

It has been continuously disputed on the plate tectonics of the southern South China block in the Early Mesozoic (c.f., Hsü et al., 1990; Carter and Clift, 2008; Cai et al., 2008). One group of authors hold the idea of a NW–SE convergent orogeny across this region derived either from the collision between the Yangtze and Huanan blocks in the Triassic (Hsü et al., 1988, 1990) or the subduction of the Paleo-Pacific plate beneath the South China block (250–190 Ma; Li and Li, 2007). In contrast, other authors prefer a dominant NNE–SSW or NE–SW convergent orogeny that was attributed to the collision between the Indochina and South China blocks (250–230 Ma, Lepvrier et al., 2004, 2011; Zhou et al., 2006; Cai and Zhang, 2009). In addition, a mixed model also was proposed as the coexistence of the NW-directed subduction of the Paleo-Pacific plate beneath the South China block and the NE–SW tectonic activation of the already amalgamated Indochina and South China blocks (Carter et al., 2001; Carter and Clift, 2008). In all of these tectonic models, a foreland basin, either of a peripheral (e.g., Lehrmann et al., 2005, 2007; Cai and Zhang, 2009) or retroarc (e.g., Li and Li, 2007) setting, called the Nanpanjiang Basin, was described, due to its development of a huge succession of deep-water siliciclastic rocks in the Middle Triassic (245–228 Ma) (GXGMR, 1985; GZGMR, 1987; YNGMR, 1990). These clastic deposits were regarded as orogenic origin (e.g., Hsü et al., 1990; Li and Li, 2007; Carter and Clift, 2008; Cai and Zhang, 2009), and, therefore, tracing their provenance may help to clarify the Early Mesozoic geodynamics in this part of the South China plate.

In the past, the deposits were most commonly interpreted to be products from recycled orogens nearby in terms of their detrital framework mode, comprising primarily angular quartz and lithic fragments with extremely low feldspar contents (e.g., Gou, 1985; Xia et al., 1993; Ceng et al., 1995; Chen et al., 2003; Goers et al.,

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2012). However, in detail there are various proposals as to the nature and location of the orogens. For example, limited geochemical data from clastic rocks have been used to characterize the orogen as either active margin, island arc (Xia et al., 1993; Chen et al., 2003) or uplifted passive margin (Mou et al., 1990; Zhang et al., 1997). Furthermore, facies models (e.g., Bouma sequence and submarine fan) and sequence stratigraphy along with various ratios of sandstone to mudstone beds as well as grain-size variation were routinely invoked to trace the location of the orogens by patterning the distribution trend of these clastic deposits basin wide (c.f., Gou, 1985; GXGMR, 1985; Ceng et al., 1995; Cai et al., 2008; Goers et al., 2012). These lines of evidence, when coupled with the paleocurrent information extracted from sedimentary structures such as cross-bedding and sole marks, however, have concluded various orogenic sources in the south (GXGMR, 1985; Cai et al., 2008), southeast (Xia et al., 1993), east (Gou, 1985; Ceng et al., 1995), or northeast (Chaikin, 2004) of the basin, or some blended sources (Mou et al., 1990; Chen et al., 2003; Goers et al., 2012; Wood et al., 2012). The arrival time of the Middle Triassic siliciclastic flux in the basin was also used to trace the source area in the south (Lehrmann et al., 2005, 2007). These diversified interpretations add further complexes to those plate tectonic models mentioned above (Carter and Clift, 2008; Cai et al., 2008), and therefore, the controversies seem to be continuing.

In spite of these disputes, previous studies did recognize the important role played by paleocurrent orientations in tracing the potential orogenic sources of this suite of flysch deposits (e.g., Gou, 1985; Xia et al., 1993; Ceng et al., 1995; Lehrmann et al., 2005; Carter and Clift, 2008; Goers et al., 2012; Wood et al., 2012). Nevertheless, in practice, the sparsity of paleocurrent indicators in the field often hinders the application of this method. As an alternative approach, magnetic fabric (e.g., anisotropy of magnetic susceptibility, AMS) analysis is a relatively well-founded method in determining grain shape fabrics of clastic rocks, and thus providing paleoflow direction information (Rees, 1961, 1965; Hamilton and Rees, 1970; Von Rad, 1970; Channell et al., 1979; Taira and Scholle, 1979; Piper et al., 1996; Baas et al., 2007; Lenhardt et al., 2013). Advantage of this method is that it can be applied to clastic rocks with no apparent depositional structures, particularly paleocurrent indicators, which are not often available for the majority of deep-sea deposits. In addition, if paleomagnetic direction can be retrieved, it can be applied to correct for potential block rotations experienced by the stratigraphic units. This is particularly useful when we deal with source regions that are located outside of blocks where clastic sediments finally settle down. Thus, a true paleoflow pattern can be obtained.

Here we report our recent study of paleocurrent directions through paleomagnetic and magnetic fabric analysis on the fine sandstones, and siltstones collected from a middle Triassic transection located in the east-central part of the Nanpanjiang Basin, South China, in an attempt to trace the origin of these clastics. Rationale for the use of this stratigraphic cross-section, combined with paleomagnetic data, is its potential for pinpointing the possible source regions, because it resides in a position with essentially equal distance to several uplifted blocks surrounding the basin that were frequently described in previous studies (e.g., Gou, 1985; Xia et al., 1993; Ceng et al., 1995; Chen et al., 2003; Cai et al., 2008; Carter and Clift, 2008; Cai and Zhang, 2009; Goers et al., 2012; Wood et al., 2012). The results indicate that most samples have preserved the primary depositional magnetic fabrics that reveal an interesting pattern of paleoflows. When corrected for magnetic declination and recovered for paleolatitude, a tectonic model of paleoplate reconstruction has been proposed thereof, which may be helpful in understanding the Early Mesozoic geodynamics of the southern South China block as well as the plate tectonics of the contemporary southeastern Asia.

2. Geological setting

2.1. General outline of the Nanpanjiang Basin

The Nanpanjiang Basin with a relict area of ca. 250,000 km^2 is located in the southern part of the South China block (Fig. 1A) and is adjoined to the south with the Indochina block through the Dian-Gui suture (Wu et al., 1999; Cai and Zhang, 2009). Several tectonic uplifts surround the basin, such as the Jiangnan oldland, the Kangdian massif, the Yunkai massif, and the northern Vietnam terrane (Fig. 1B). Among these, the former three units expose mainly continental basement, representing metamorphic rocks and enclaves of Caledonian (413–465 Ma) or earlier granitoids (GXGMR, 1985; GZGMR, 1987; GDGMR, 1988; YNGMR, 1990), whereas the latter one is almost covered by Paleozoic sedimentary sequences (Lepvrier et al., 2011). These uplifted areas also control the regional tectonic lineaments that show a systematic transition in strike from W to E from NE–SW through nearly E–W and WNW–ESE to again NE–SW.

The basinal region has developed three stratigraphic successions separated by two angular unconformities residing between the Lower and Upper Paleozoic and between the Middle and Upper Triassic, respectively, witnessing a relatively complex geological evolution (GXGMR, 1985; GZGMR, 1987; GDGMR, 1988; YNGMR, 1990). In the Early Paleozoic, the basinal region, together with the remaining part of the South China plate, was dominated by a neritic to hemipelagic setting until the Silurian when the Caledonian orogeny uplifted this region (Guo et al., 1989; Ren and Chen, 1989). Subsequently, a shallow sea environment occurred from the Early to latest Middle Devonian (GXGMR, 1985), which was succeeded by a deep-sea setting through much of the late Paleozoic with deposition of mudstone, shale, and intercalated siliceous rocks (Wang, 1994; Wang et al., 1995; Wu et al., 1994). Concurrent with this deep-sea setting was the development of several isolated shallow-water platforms within the basin region (DQGPG, 1992; Lehrmann et al., 2005, 2007) and sporadic eruptions of mafic to ultra-mafic volcanics in its central and southern parts (c.f., Cai and Zhang, 2009; Yang et al., 2012). Since the Late Permian, Indosinian orogeny (P_2-T) began to influence the basin area with a brief deposition of bauxite horizons in the northern Vietnam terrane (Lepvrier et al., 2011) and molasses in the Shiwandashan area (Liang and Li, 2005) in the earliest Late Permian. This orogeny was accompanied by amalgamation of several continental blocks (such as North China, South China, Indochina, Sibumasu) to form the southeastern Asian part of the Eurasian continent (e.g., Sengor et al., 1988; Hutchison, 1989, 1993; Metcalfe, 2002). From the late Early throughout Middle Triassic, this basin was filled by voluminous siliciclastic debris through sedimentary gravity-flow deposition (GXGMR, 1985; GZGMR, 1987; YNGMR, 1990). By the Late Triassic, the basin was essentially uplifted by the Indosinian (P_2-T) orogeny, resulting in deposition of shallow-sea coarse clastic rocks along its western margin (YNGMR, 1990), marine molasses in the Shiwandashan area (GXGMR, 1985), and braided-stream molasses along its northern margin (Enos et al., 1998). Molasses deposition of alternating marine and continental facies prevailed in its southeastern part until the Early Jurassic (GXGMR, 1985), but after that, the basin, accompanying the entire South China block, ended its long-term marine history since the late Paleozoic.

2.2. Stratigraphy of the Middle Triassic

In most part of the Nanpanjiang Basin, this suite of Middle Triassic siliciclastic rocks is marked at its base by a thin layer of acidic tuffs or tuffaceous clastic rocks and is terminated by an unconformity contact with the Upper Triassic or younger formations (Fig. 2)

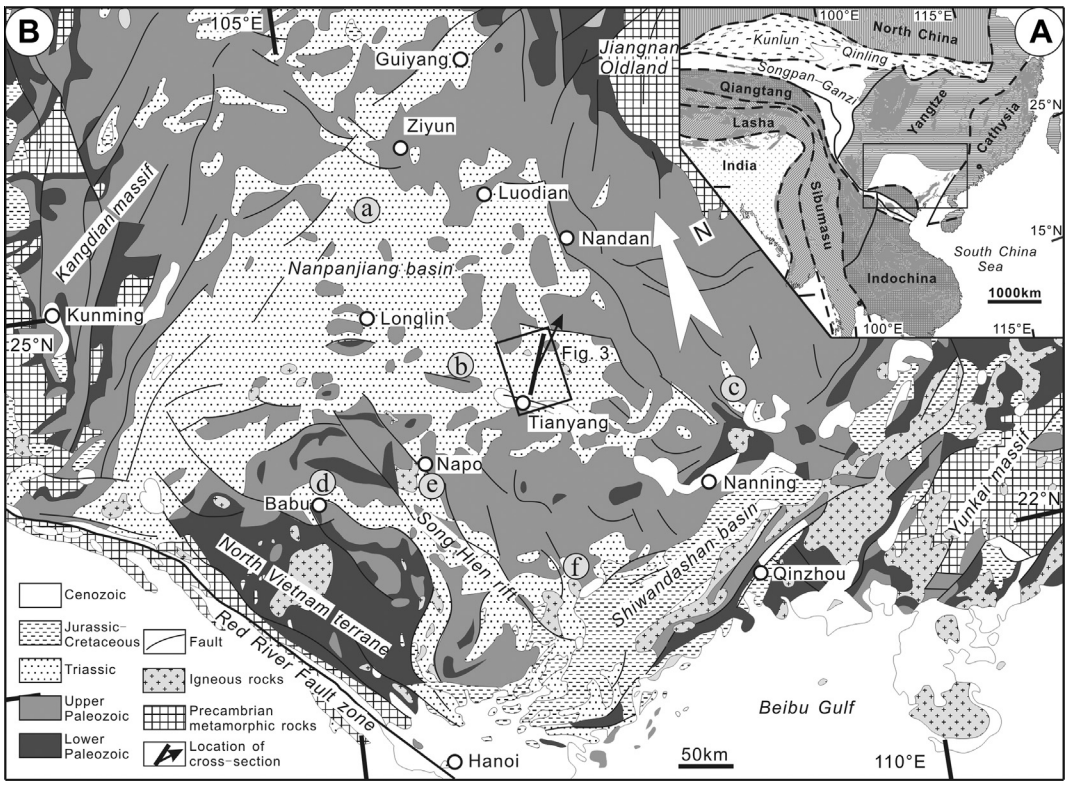


Fig. 1. (A) Outline of Southeast Asia with insert box showing the location of B). (B) Sketch map of the Nanpanjiang Basin and adjacent areas with the thick line denoting the location of the Middle Triassic cross-section studied. The arrow indicates the mean paleocurrent orientation inferred from this study. Letters a–f denote the location of the stratigraphic columns in Fig. 2 and the box shows the location of Fig. 3. Redrawn after CAGS (1975, 1997), GXGMR (1985), GZGMR (1987), GDGMR (1988) and YNGMR (1990).

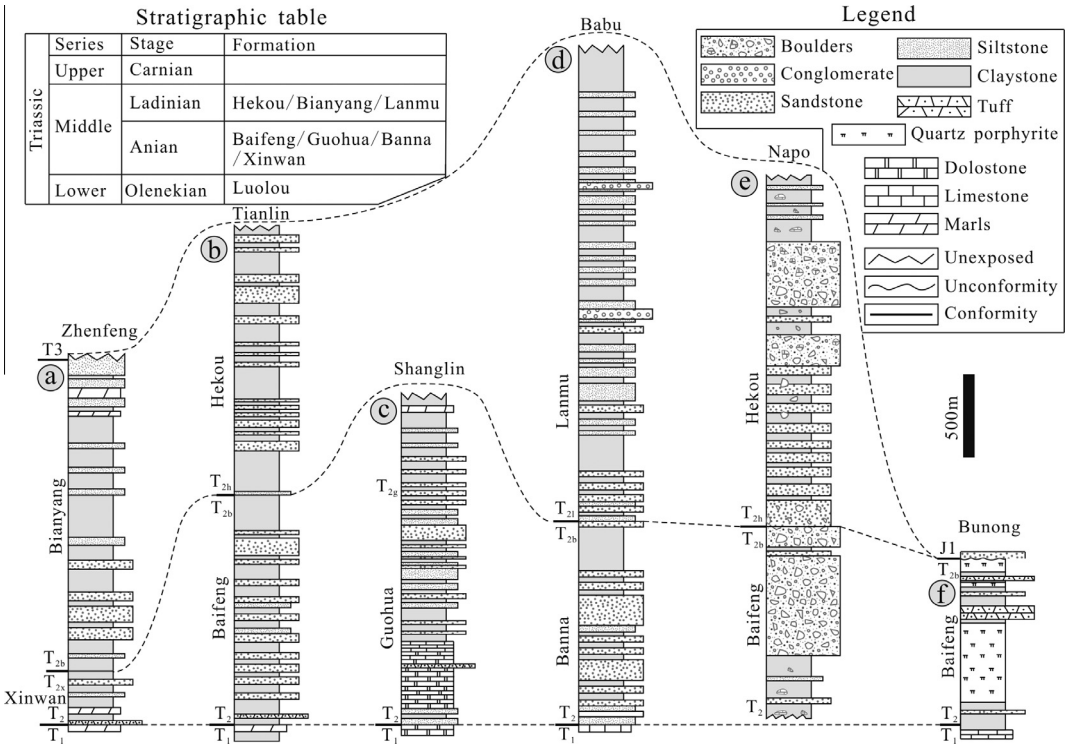


Fig. 2. Stratigraphic columns distributed across the Nanpanjiang Basin. Column a is redrawn after GZGMR (1987), columns b, c, e and f after GXGMR (1985), and column d after Huang (1998). See Fig. 1B for the location of these columns in the Nanpanjiang Basin.

(GXGMR, 1985). Interbedded sandstone, siltstone and mudstone dominate this sequence with local occurrence of conglomerates, and the relict thicknesses are most between 3.0 and 5.0 km with a maximum of up to 9.0 km (GXGMR, 1985; Zhang, 1986, 1988).

Two formations have been identified from this siliciclastic sequence, up section, i.e. the Baifeng (T_2b) (Guohua, Banna and Xinwan) and Hekou (T_2h) (Bainyang and Lanmu) Formations (Fig. 2) (GXGMR, 1985; GZGMR, 1987; YNGMR, 1990; Huang, 1998). The Baifeng Formation is relatively widespread within the basin with a maximum relict thickness of 6484 m. The base and top parts of this formation primarily comprise mudstone intercalated with fine sandstone and siltstone, whereas the remaining part is dominated by thick-bedded fine lithic sandstone. In contrast, the Hekou Formation is restricted in distribution mainly to the central part of some synclines and has relict thicknesses up to 3129 m. Mudstone dominates this formation with subordinate fine sandstone and siltstone. These lithologies form several secondary fining-up depositional cycles with sandstone at the bottom grading upwards to mudstone.

2.3. Lithofacies

A gradual change in lithofacies from slope to basin setting is represented by the Middle Triassic deposits across the Nanpanjiang Basin from south to north (Fig. 2).

On the southern margin of the basin, slope deposits occur as the coarse lithologies (e.g., conglomerates) of slid, slump or avalanche origin (Fig. 2d). In some cases, they form huge intervals, such as the 1432 m thick, poorly sorted, polygenetic clay boulders of the top Baifeng to middle Hekou Formations in the line from Napo to Babu cities (Figs. 1B and 2e) (GXGMR, 1985; Huang, 1998). These coarse lithologies continue southwards into the so called "Song Hien rift" in the border area between China and Vietnam where they also contain variously sized blocks of marbles and basalts and extend even into the Lower Triassic (e.g., Lepvrier et al., 2011). Turbiditic flow deposits with the typical Bouma Sequence (Ta-Te) (Bouma, 1962) frequently occurs on this margin and show sedimentary structures, such as cross-lamination, convolute bedding, grading and sole casts (GXGMR, 1985).

Northwards, the lithofacies gradually transit to base-of-slope or basin-floor deposits. Fine-grained lithologies (i.e., fine sandstone, siltstone and mudstone) prevail for this main part of the basin (Fig. 2b and c), which are increasingly notable towards its northeastern margin, such as in the cities of Nandan and Ziyun where mudstone and subordinate siltstone dominate the Middle Triassic rocks (Figs. 1B and 2a) (GXGMR, 1985; GZGMR, 1987). These fine lithologies mainly appear as moderately or thickly bedded massive sand bodies with sheet-like geometries and occasionally with escape pipes, characteristic of debris-flow deposits of Shanmugam (2000). The turbiditic flow deposits locally occur, but becomes increasingly incomplete in the Bouma Sequence from short of the basal conglomerates of the graded division Ta north of Napo, through absent of the whole Ta division in the central part of the basin, to remaining of only the two top divisions, Td and Te, close to the northern margin of the basin (GXGMR, 1985).

2.4. Lithology

For the studied cross-section, sheet-like fine sandstone, siltstone and mudstone are the dominant lithology with single bed thicknesses ranging from centimeters to meters and now are preserved in a series of open folds trending generally NW–SE (Figs. 3 and 4a). Sedimentary structures, such as sole casts (Fig. 4b) and cross bedding (Fig. 4c), occur locally, but most strata appear as missive sands. Microscopic observations of the clastic rocks from the studied cross-section show no trace of deformation and metamorphism or any phenomenon related to hydrothermal percolation (Fig. 4d).

These fine deposits have a fairly uniform detrital framework mode, characterized by low textural but relatively high compositional maturity. The former feature is illustrated by angular outlines of framework grains with long dimension generally less than 0.2 mm (Fig. 4d). The latter characteristic is exemplified by the relatively monotonic detrital composition rich in quartz (37–54 Vol.%) and lithic particles (21–40 Vol.%) and poor in feldspar (generally less than 3 Vol.%), particularly potassic feldspar (Cai, 2009). Sedimentary debris, like carbonate and chert, is common especially towards the north of the basin, whereas volcanic and metamorphic fragments are generally low in content and constitute the primary paramagnetic minerals except for clays, such as muscovite, biotite and other silicates. Comparably, matrix is relatively abundant, varying between 14% and 33% by volume, and consists of clays, carbonates, very fine lithic particles and minor Fe–Ti oxides or Fe sulfides. Framework grains are cemented by clays or carbonates, of which carbonate cementation is more common towards the north part of the basin.

These general petrographic characteristics dominate the central and northern parts of the basin (e.g., GXGMR, 1985; Gou, 1985; Ceng et al., 1995; Cai, 2009). However, the petrographies gradually change, when approaching the southern margin of the basin where feldspar and volcanic lithics increase evidently in content (GXGMR, 1985).

3. Methods

This study selected a lithologic cross-section through the Middle Triassic along the road linking Tianyang and Bama cities in the central-eastern part of the basin (Figs. 1B and 3A). The extent of weathering is generally low such that fresh outcrops are wealthy to sample. Sampling was made mainly in the Baifeng Formaton, while, due to poor exposure, the Hekou Formation was sampled only at one site (Fig. 3B). The collected samples are primarily from massive fine sandstone and siltstone. A total of 262 cores were drilled in 32 sites using a portable gasoline-powered drill, and oriented using a pomeroy orientation device and a magnetic compass. The cores collected from the field were individually cut in the laboratory into 2–3 specimens with diameter 2.5 cm and length 2.2 cm. After that, 518 specimens were selected for NRM (natural remanent magnetization) measurement, 60 for thermal demagnetization, 29 for alternating field (AF) demagnetization and 120 for AMS measurement. All data sets for this study are available in the online data repository.

Step demagnetization was performed in the shield room at the Paleomagnetism Laboratory of the South China Sea Institute of Oceanology, Chinese Academy of Sciences, which has a relict ambient field less than 300 nT. Thermal demagnetization was conducted using the Schonstedt TSD-1 demagnetizers with temperatures set at: NRM, 50, 100, 125, 150, 200, 250, 300, 330, 360, 390, 420, 440, 460, 480, 500, 520, 540, 560 and 580 °C. At individual temperatures, samples were heated ca. 40 min and then removed to the cooling region of the oven. AF demagnetization was done with a three-axis 2G demagnetizer at incremental fields of NRM, 2, 4, 6, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90 and 100 mT. Demagnetized samples were placed in a mu-metal shielded cylinder before remanence measurement. Ambient fields in the heating oven and its cooling region are less than 10 nT, and the shield cylinder less than 5 nT. Remanence measurement was made using a 2G three-axis cryogenic magnetometer, which has a measurement lower limit of 10^{-12} A m^2. The resulting demagnetization data were processed through principal component analysis (Kirschvink, 1980) and Fisher statistics (Fisher, 1953) in Enkin's PMGSC, and the acquired ChRM directions are checked using F fold test (McElhinny, 1964) and incremental fold test (Watson and Enkin, 1993). The samples that either have no

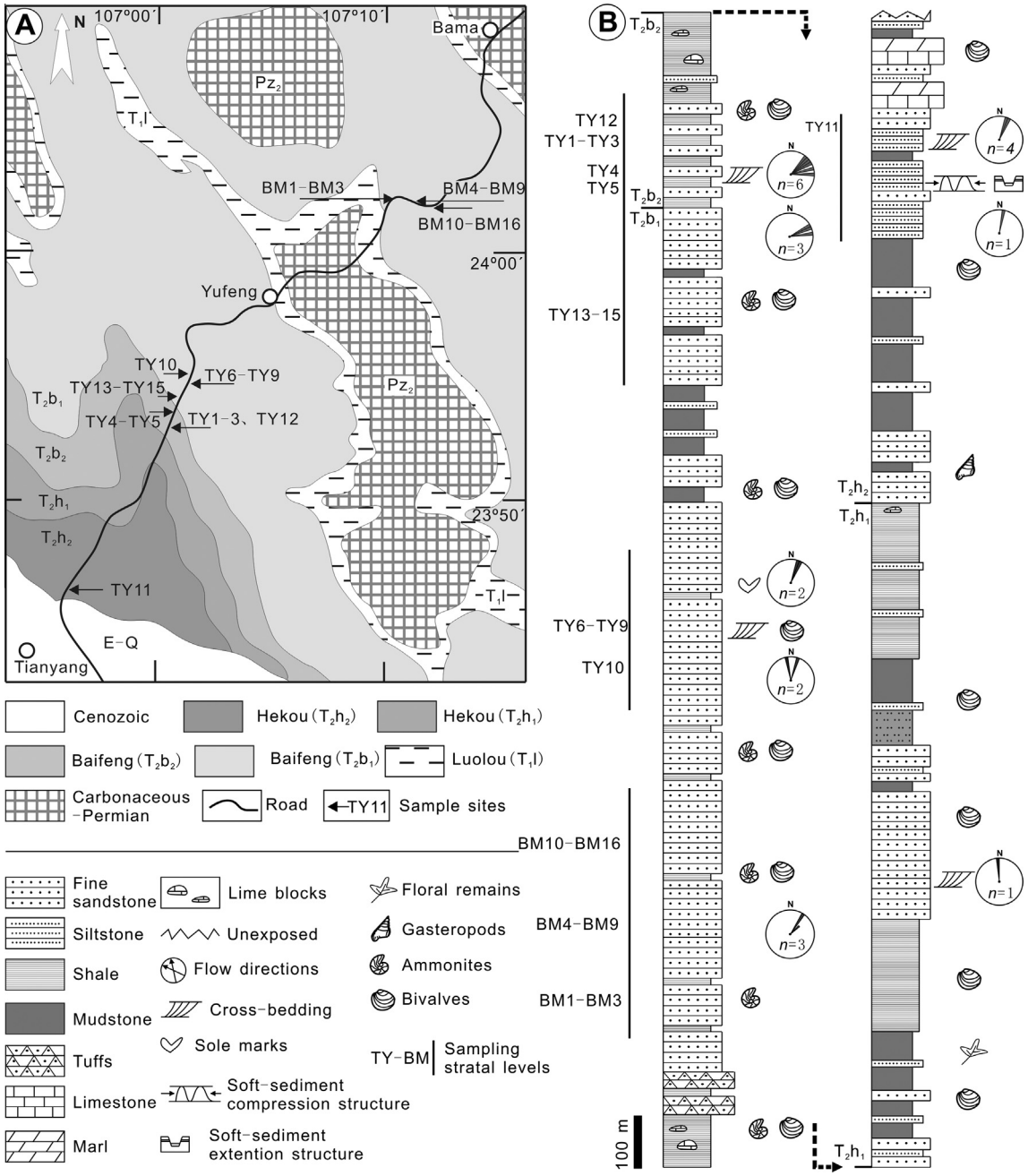


Fig. 3. (A) Geological map of the studied section showing the sampled sites (black arrows) along the road connecting the Tianyang and Bama cities. See Fig. 1B for location of this map. (B) Stratigraphic column in the studied section. Redrawn after GXGMR (1985). Sampling sites are shown on the left side of the column, and the marks of paleoflows, fossils, and syn-sedimentary structures shown on the right side. The flow directions (in Rose diagram) from sedimentary structures are shown on the right side of the corresponding marks, and the others are from AMS analysis. See Supplementary Table A1 for the details of sampling information and Tables A4 and A5 for paleocurrent data illustrated.

reliable ChRMs or have a large scatter in directional distribution are discarded.

Low field AMS (anisotropy of magnetic susceptibility) was measured at room temperature using a MFK1 V4.0 kappa bridge with a field of 400 A/m and a frequency of 976 Hz at the Environmental Magnetism Laboratory of the Guangzhou Institute of Geochemistry, Chinese Academy of Sciences. Under these conditions, the precision of the magnetic susceptibility meter can reach 2×10^{-8} SI. The resulting analytical data were processed by Jelinek statistics (Jelinek and Kropacek, 1978) in Anisoft 42.

AMS can be used as a proxy of grain shape fabrics, the statistic arrangement of elongate mineral grains, in sedimentary rocks that is governed by flow regimes (c.f., Baas et al., 2007). Many studies were involved in recovering paleoflow directions through AMS method (e.g., Rees, 1961, 1965; Hamilton and Rees, 1970; Von Rad, 1970; Channell et al., 1979; Taira and Scholle, 1979; Piper et al., 1996; Veloso et al., 2007; LenHardt et al., 2013). In contrast to traditional methods that heavily rely on flow indicators, such as cross-bedding, sole casts, and gravel clasts, AMS method can be used for strata without sedimentary structures and, therefore, is particularly suitable for massive debris-flow deposits. Baas et al. (2007) gave a comprehensive review of the general consistence between paleoflow directions recovered from clastic sequences worldwide by sedimentary structure and AMS methods,

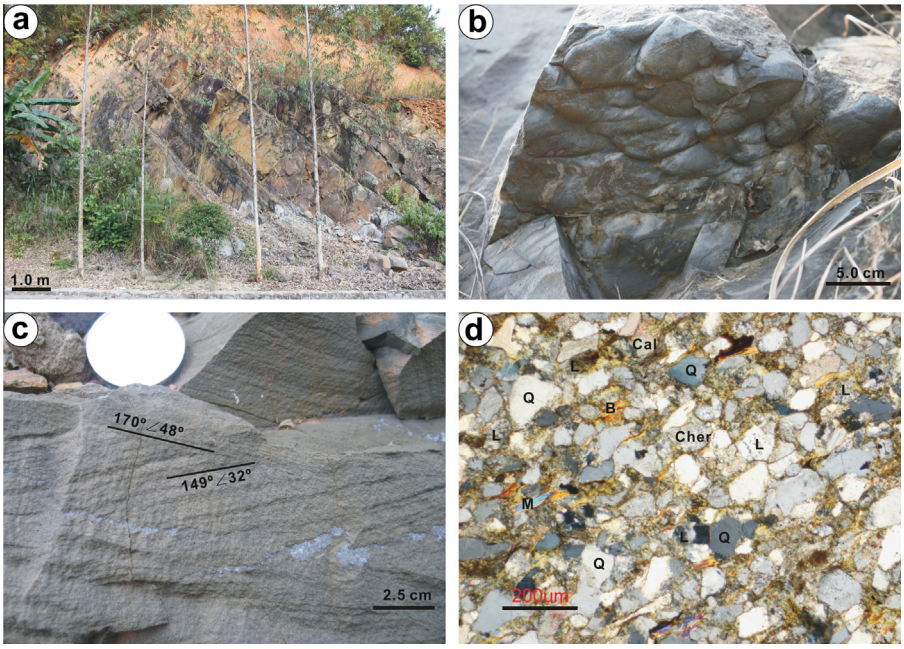


Fig. 4. Field and microscopic photos. (a) Massive thick-bedded fine sandstone at site TY10. (b) Sole casts at site TY7. (c) Cross-bedding with in situ attitudes (149°∠32°) (dip direction ∠dip angle) in a sandstone bed (170°∠48°) at sites TY11, which corresponds to a N23°E paleocurrent direction after untilting. (d) Microscopic image of fine sandstone with preferred orientation of detrital grains at site TY11. Thin section was prepared perpendicular to bedding. *Abbreviations:* B, biotite; Cal, calcite; Cher, chert; L, lithic fragments; M, muscovite; Q, quartz.

respectively, though controversies also exist for the massive beds, due to the lack of paleocurrent indicators for comparison. Therefore, it is necessary to test on the AMS-derived flow directions using flow indicators when available.

Here, AMS-derived flow directions from the massive strata are tested using those recovered by the flow indicators from the neighboring beds at the same sites. The results show a generally good consistence, which constitutes the basis for the interpretation of AMS results from the massive beds without flow indicators available in the neighboring strata. In addition, for the latter case, comparison is also made using previous results from the basically same stratigraphic levels but in the neighboring areas. In the worst case where sedimentary structures are not available from present and previous studies, the AMS-derived flow directions from multiple stratigraphic levels at the adjacent sites could be simply compared on the reasoning that random deviations from flow directions should not be consistent across a significant stratal interval.

4. Paleomagnetic results

Natural remanent magnetization (NRM) intensities measured for 518 samples are basically in the order of 10^{-4} A/m with an average of $4.25 \pm 1.77 \times 10^{-4}$ A/m. On a histogram plot (Fig. 5a), NRM shows a single-modal, normal distribution. Among these, 89 samples were demagnetized in order to correct for potential local, regional and/or block rotations experienced by paleocurrent directions after lock-in during diagenesis. Characteristic remanence components, ChRM, can be readily isolated at temperatures greater than 500 °C, and at magnetic fields larger than 20 mT, possibly indicating that magnetite is primary remanence carrier. The in situ NRM directions are distributed with an average of 89 samples at $D = 0.5°$, $I = 46.4°$, $\kappa = 15.2$, $\alpha_{95} = 4.0°$, close to the expected present magnetic field direction ($D = 358°$, $I = 36°$). Of these, 55 samples retain the relatively well-defined ChRM, including 18 AF and 37 thermal demagnetized samples. Representative demagnetization curves refer to Fig. 6. The resulting mean ChRM directions of the five localities in the Baifeng Formation in geographic coordinates are distinguished from the modern magnetic field direction (Fig. 7c and Supplementary Table A2). No reliable direction was acquired for the samples gathered from the only one site in the Hekou Formation, due to the small number of samples analyzed.

The sampled strata occur in a series of folds, thus enabling fold tests on the ChRM directions acquired. For the Fisher means of the five localities in the Baifeng Formation (Fig. 7c and d), complete unfolding causes a significant improvement in clustering with the ratio of precision parameter, $\kappa_s/\kappa_g = 10.05$ (where subscripts s and g refer to stratigraphic and geographic coordinates, respectively), larger than the corresponding F value 6.03 at 95% confidence level (McElhinny, 1964) and thus they pass the F fold test. Incremental fold test (Watson and Enkin, 1993) achieves a similar conclusion by demonstrating that an optimum degree of untilting occurs at 108 ± 18.5% unfolding (Fig. 8a) with a mean paleomagnetic direction, $D = 46.0°$, $I = 13.4°$, $\kappa = 41.9$, $\alpha_{95} = 12.0°$, close to the one at 100% unfolding, $D = 45.4°$, $I = 14.8°$, $\kappa = 39.2$, $\alpha_{95} = 12.4°$. For Fisher mean direction of 55 samples (Fig. 7a and b), the ratio of precision parameter, $\kappa_s/\kappa_g = 2.49$, is greater than the F value 1.6 at 95% confidence level (McElhinny, 1964). Therefore, complete unfolding also results in a significant increase in clustering of the ChRM directions of individual samples. This positive fold test is further supported by incremental fold test (Watson and Enkin, 1993) that reaches an optimum degree of untilting at 89.2 ± 2.2% unfolding (Fig. 8b). The corresponding mean paleomagnetic direction ($D = 43.5°$, $I = 19.2°$, $\kappa = 9.96$, $\alpha_{95} = 6.4°$) is also close to that at 100% unfolding ($D = 44.8°$, $I = 16.9°$, $\kappa = 9.7$, $\alpha_{95} = 6.\ 5°$). Therefore, both F test and incremental fold test indicate that ChRMs were obtained prior to folding.

This study takes the Fisher mean ($D = 44.8°$, $I = 16.9°$, $\kappa = 9.7$, $\alpha_{95} = 6.5°$) (Fig. 7b) of 55 samples as the average paleomagnetic direction of the Middle Triassic section. This direction is close at 95% confidence to the one ($D = 55.9°$, $I = 8.3°$, $\kappa = 8.3$, $\alpha_{95} = 13.3°$) obtained by Gilder et al. (1995) in the same location. The individual paleomagnetic directions for the five localities in the Baifeng Formation depart this mean direction by no more than 8° (Fig. 7d),

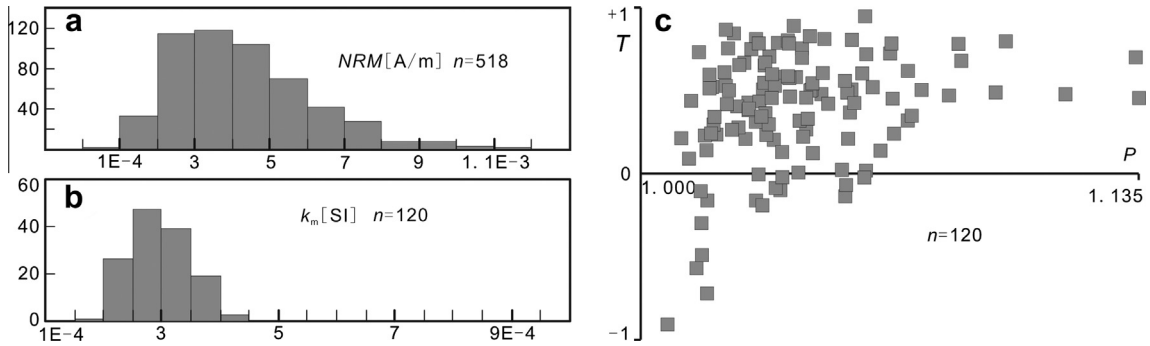


Fig. 5. AMS and remanence parameters. (a and b) Histograms of natural remanent magnetization (*NRM*) and bulk magnetic susceptibility (k_m), respectively. (c) *T* (shape factor) versus *P* (degree of anisotropy) plot. See the online files "Sus. and NRM" and AMS for data used.

which may be caused by the inability in averaging out the secular variations of the paleomagnetic field, but not by the local rotation of strata in the studied cross-section. The corresponding Middle Triassic paleomagnetic pole (N44.6°, E209.5°, $dp/dm = 3.5°/6.7°$) is essentially identical to the mean Early-Middle Triassic paleomagnetic pole (N43°, E217°, $\alpha_{95} = 2.8°$) for the South China block (Fig. 7e) (Gilder et al., 2008), thus excluding a significant regional rotation of the studied cross-section relative to the South China block. Thus, this mean direction corresponds to a paleolatitude ca. N8.6° and a clockwise block rotation of ca. 45° for the studied cross-section together the South China block as a whole.

5. AMS analysis

5.1. Currents and AMS fabrics

Present knowledge on the relationship between currents and magnetic fabrics will help us interpret the AMS results. The preferred orientation of elongate magnetic grains on a sediment bed by currents immediately before deposition forms several common fabric patterns. In still waters, particles settle through the water column to come to rest with their maximum flatten plane lying in the sedimentary bed without apparent orientation. In such a situation, minimum axes (k_{min}) of AMS ellipsoid of populations of magnetic grains will cluster in the direction perpendicular to the bedding plane, while intermediate (k_{int}) and maximum (k_{max}) axes will lie randomly in the bedding plane to form a bedding-parallel magnetic foliation (e.g., Tarling and Hrouda, 1993; Baas et al., 2007). Under moderate or weak currents, grain long axes tend to be flow aligned, but to be imbricated upstream, causing the short axes to deviate from the pole of bedding (Harms et al., 1982; Allen, 1984). Accordingly, the magnetic foliation comprising k_{max} and k_{int} will lean by a small angle away from the bedding plane and the dip of grouped k_{min} will determine the flow direction (e.g., Tarling and Hrouda, 1993; Baas et al., 2007; Veloso et al., 2007). With increasing current velocity, grain long axes will be arranged perpendicular to flow direction, while intermediate axes tend to be imbricated upstream (Harms et al., 1982) or intermediate and short axes are randomly oriented in a plane parallel to flow and perpendicular to bedding (Ellwood, 1980). In such cases, the k_{max} will cluster to form a magnetic lineation oriented perpendicular to the flow direction (Baas et al., 2007).

5.2. AMS results

For all 120 samples analyzed, the bulk susceptibility values are essentially in the order of 10^{-4} SI, varying between 1.31–3.74×10^{-4} SI with a mean of $2.45 \pm 0.54 \times 10^{-4}$ SI. Like NRM, a histogram plot of bulk susceptibility also shows a single-modal normal distribution (Fig. 5b), suggesting that the magnetic components of sandstone and siltstone are relatively constant. Most of samples show oblate fabrics with shape factor $T > 0$, and degree of anisotropy between 1.0 and 1.1 (Fig. 5c), indicating a dominant fabric type of magnetic foliation. The mean AMS parameters, such as degree of lineation, degree of foliation, degree of anisotropy and shape factor, for samples from various sites are listed in Supplementary Table A3. Of these, the samples (TY6–TY10) from the middle part of the Baifeng Formation (T_2b) have the highest shape factor ($T = 0.584–0.897$) and the lowest degree of lineation ($L = 1.001–1.006$).

The fabric patterns from AMS analysis on samples cored in the lower part of the Baifeng Formation (Fig. 3B) reveal an overall NE-directed flow (Fig. 9a–c). Samples from sites BM1 to BM3 were collected from the basal portion of the section. In geographic coordinates, magnetic foliation comprising k_{max} and k_{int} axes and grouped k_{min} axes lies close to bedding planes and poles, respectively (Fig. 9a), characteristic of sedimentary fabrics; after bedding-tilt correction (i.e., in stratigraphic coordinates) (Fig. 9a′), k_{min} and k_{int} tend to form a weak girdle perpendicular to magnetic lineation comprising k_{max} axes, indicating a SW–NE oriented stronger flow. Samples from sites BM4–BM9 show a similar fabric pattern, but the k_{min} axes are slightly deviated to the NE from poles of bedding planes (Fig. 9b), thus further constraining the flow towards NE. This flow direction is best represented by fabric pattern of the samples from sites BM10–BM16 whereby k_{min} axes in stratigraphic coordinates are notably deviated from vertical direction to NE (Fig. 9c′).

In contrast, an obviously different current regime (Fig. 9d and e) is recorded by the samples cored in the middle part of the Baifeng Formation (Fig. 3B). Samples from site TY10 have a scatter distribution of magnetic susceptibility principal axes (Fig. 9d and d′). In geographic coordinates, k_{max} and k_{int} form a magnetic foliation more or less parallel to the bedding planes, and k_{min} axes tend to incline to NE away from bedding poles (Fig. 9d). After 100% tilt correction, the NE deviation of k_{min} axes remains (Fig. 9d′), but there is no clear clustering for either k_{max} or k_{int} axes because of a limited amount of samples. This is overcome by the relatively large number set of samples from sites TY6–TY9 that exhibit a much better fabric pattern. For these samples, the clustering of k_{max}, k_{int} and k_{min} is apparent in both geographic and stratigraphic coordinates (Fig. 9e and e′). In stratigraphic coordinates, for example, the magnetic foliation inclines from the horizontal plane by a much larger angle, while k_{min} axes deviate from the vertical direction dipping to NNW (Fig. 9e′), thus suggesting a NNW directed, weak flow. Such a weak flow may in part contribute to the scatter in distribution of grain fabrics (Fig. 9d′ and e′).

The upper part of the Baifeng Formation registers NE-oriented flows (Fig. 3B; Fig. 9f–h). For example, samples from sites TY13–TY15 in geographic coordinates show a typical sedimentary fabric

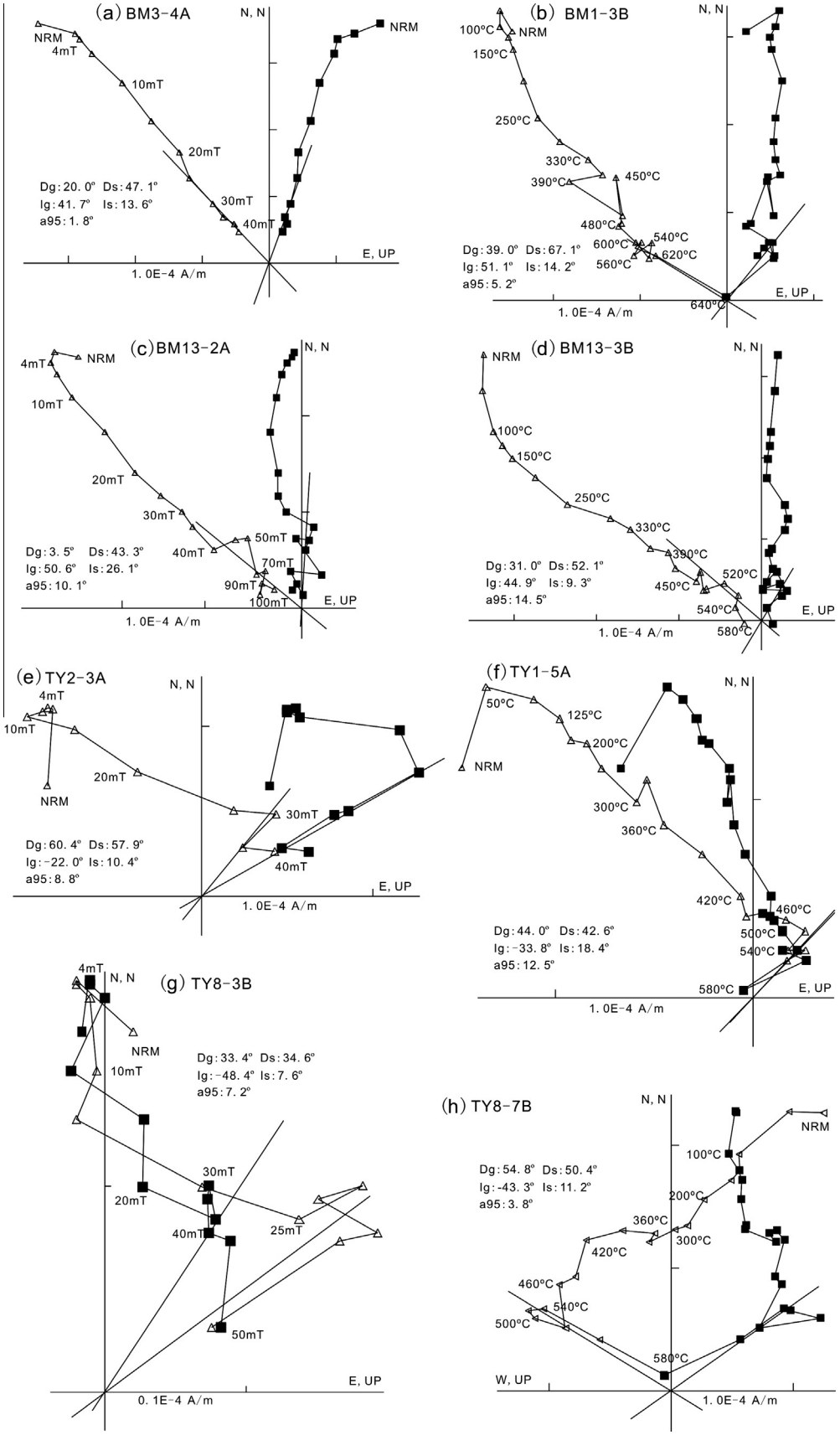


Fig. 6. Representative curves (orthogonal projection) of stepwise AF (left panel) and thermal (right panel) demagnetization. An extensive data set of demagnetization is available in the online file "Demagnetization-individual samples".

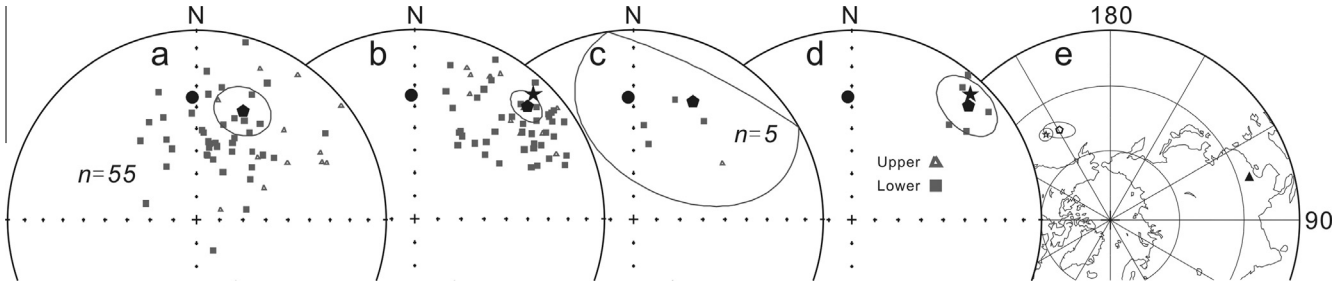


Fig. 7. Characteristic remanence directions and paleomagnetic poles (equal-area projection). (a and b) Directions for all 55 samples in geographic and stratigraphic coordinates, respectively. (c and d) Mean directions for five localities from the Baifeng Formation in geographic and stratigraphic coordinates, respectively. Data is available in the online files "Demagnetization-55 and -means". (e) Early to Middle Triassic paleomagnetic poles for the South China block. Filled triangle stands for the sampling location (N24°, E107.5°) and black circles for present magnetic field direction in study section. Black and hollow pentagons denote the mean magnetic field direction and the corresponding paleomagnetic pole, respectively. Hollow and black stars represent the mean Early-Middle Triassic paleomagnetic pole and the reference direction expected from this pole (Gilder et al., 2008).

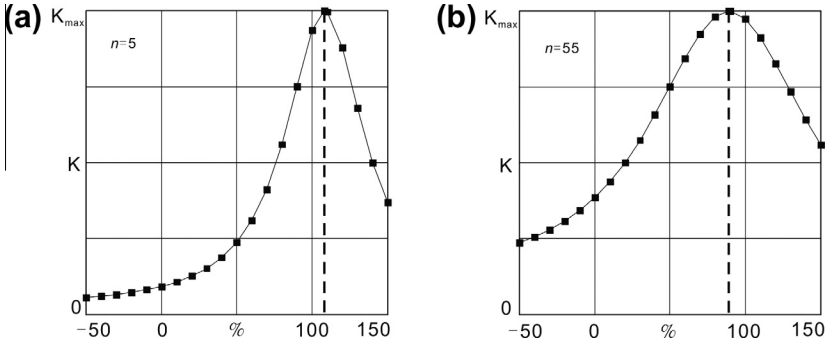


Fig. 8. Incremental fold tests (Watson and Enkin, 1993) for the characteristic remanence of (a) the means of the five localities and (b) all 55 samples.

(Fig. 9f) that, after untilting, tends to form a girdle comprising k_{min} axes orthogonal to nearly horizontal k_{max} (Fig. 9f′), thus pointing to a SW–NE flow. A similar flow regime is also obtained from sites TY4–TY5, TY1–TY3 and TY12. In geographic coordinates, the results show some extents of scatters, whereas the magnetic foliation comprising k_{max} and k_{int} axes lies near their respective bedding planes (Fig. 9g and h), indicative of a sedimentary magnetic fabric. The type of fabric becomes more apparent after untilting that renders k_{min} and k_{int} to spread along a great circle perpendicular to the nearly horizontal k_{max} (Fig. 9g′ and h′). Of these sites, the latter four further constrain the flows towards NE given the NE dipping of grouped k_{min} (Fig. 9h′). It is noteworthy that several samples at site TY12 have nearly vertical k_{max} (Fig. 9h′), which may be related to their nearly zero of shape factor.

The nearly N–S directed flow regime in the middle Baifeng Formation recurs in the upper part of the Hekou Formation (Fig. 3B). Site TY11 located in the southern limb of an anticline was from the stratal level. In geographic coordinates, k_{max} axes of the samples from this site are not aligned with the hinge of the fold, different from the deformational fabric formed during development of cleavage; k_{int}–k_{max} magnetic foliation forms a relatively large angle with the bedding planes, while k_{min} axes are deviated towards NE from bedding poles (Fig. 9i). After tilt correction, k_{min} and k_{int} tend to form a weak great circle, hence indicating a NNE-directed strong flow (Fig. 9i′). In addition, this site also contains two samples with nearly vertical k_{max}, probably due to their nearly zero of shape factor.

6. Discussion

6.1. Paleoflow pattern: field test and comparison with previous studies

The above AMS analysis indicates that the Middle Triassic rocks have preserved the primary sedimentary fabrics that present a range of flow directions from NNW to ENE with an overall orientation of NE (Fig. 10a).

These AMS-derived flow directions mainly come from the massive beds and, therefore, need to be tested using paleocurrent indicators from sedimentary structures available in the neighboring stratigraphic levels at the same sites. For the upper part of the Baifeng formation, we have recovered flow directions of N44-87°E from cross-bedding in the same location (Fig. 3B) that essentially bracket those from AMS analysis (Fig. 9f′, g′ and h′). For the upper part of the Hekou Formation, the NNE flow (Fig. 9i′) from the only one site is rather close to the N19-23°E ones from cross-bedding (Fig. 3B). The approximate N-directed flow regime seems to continue into the lower part of the Hekou Formation where a previous survey documented a N355°W-oriented flow (Fig. 3B) from cross-bedding in the studied section (GXGMR, 1985).

The consistence in recovery of flow directions between AMS and sedimentary structures can be applied to sites with no sedimentary structures available. Accordingly, the AMS analysis gets a virtually N-directed flow from the middle part of the Baifeng Formation (Fig. 9d′ and e′). This direction agrees with previous observations of flute casts in the same stratal levels in the areas immediately south of the studied cross-section (GXGMR, 1985) that revealed NNE flow directions (Fig. 3B). Furthermore, a NE flow orientation is interpreted from the lower part of the Baifeng Formation (Fig. 9a′–c′). This direction is well consistent in the three stratigraphic levels sampled and, therefore, can represent the flow regime.

Such a overall NE flow pattern is different from some previous observations that documented flow directions of dominant N in the southern part of the basin (GXGMR, 1985; Goers et al., 2012; Wood et al., 2012), NW or W in the central part of the basin (Gou, 1985; Xia et al., 1993; Ceng et al., 1995) and SW or W in the northern part of the basin (Xia et al., 1993; Chaikin, 2004; Goers et al., 2012; Wood et al., 2012) through measuring sedimentary

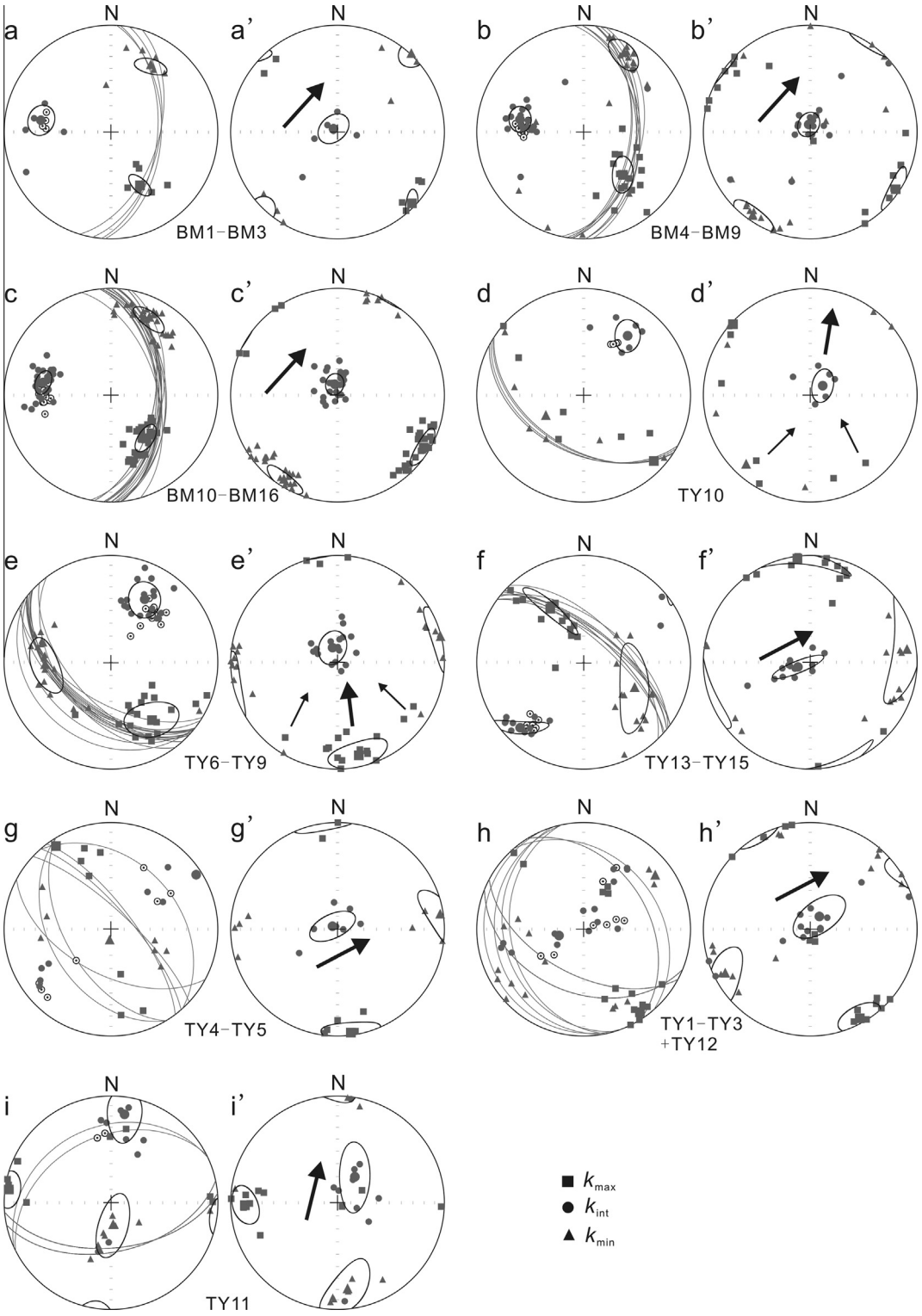


Fig. 9. AMS fabrics (lower-hemisphere, equal-area projection) in each locality in geographic (left) and stratigraphic (right) coordinates, respectively. k_{max}, k_{int} and k_{min} denote the maximum, intermediate and minimum axes of AMS ellipsoid, respectively. Circles with dot in center (⊙) stand for bedding poles and great circles for bedding planes. See the online "AMS" files for data used.

structures in this clastic sequence. Of these, the N-directed flows are common such as in the middle part of the Baifeng formation and the Hekou Formation. Yet, the SW-directed flows are not unambiguous. The AMS fabrics in several sites in the lower and upper Baifeng Formation (Fig. 9b′ and f′) can be interpreted in terms of this flow orientation, but adjacent sites (Fig. 9c′ and h′) clearly show a NE orientation and that is also verified by our field observations. Contrary to previous studies (e.g., Gou, 1985; Xia et al., 1993; Ceng et al., 1995; Carter and Clift, 2008), the flows revealed here are substantially devoid of NW or W directions. Those suspicious NW flows in the middle part of the Baifeng Formation (Fig. 9d′ and e′) are most likely the results of scattering in current

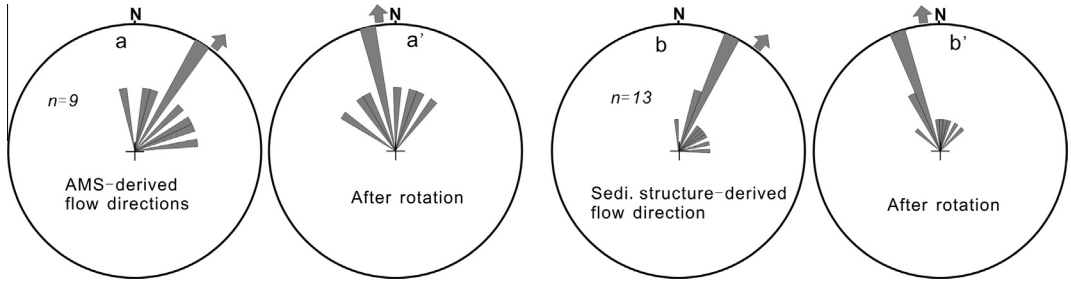


Fig. 10. Paleocurrent directions (Rose diagram) from magnetic fabrics (a and a′) and from sedimentary structures (b and b′), respectively. (a and b) Tilt-corrected paleocurrent directions; (a′ and b′) declination-corrected paleocurrent directions (paleomagnetic declination, 45°). The arrows indicate approximately the average direction. See Supplementary Tables A4 and A5 for data employed.

orientation under weak hydraulic conditions and thus may not represent the main trend. Because the sample sites in this study mainly focus on the thick-bedded sand bodies and cover multiple continuous beds, it is reasonable to conclude that the overall NE orientation may represent a primary flow regime during deposition of the Middle Triassic.

6.2. Provenance

Paleocurrent directions can be used to trace the direct subaqueous source areas given the sediments are generally transported downdip from the shelf areas. The clastic deposits in the studied cross-section appear mainly as massive sands with sheet-like geometries, more fitting to the debris-flow dominated slope or base-of-slope setting with non-channelized systems proposed by Shanmugam (2000). Accordingly, an overall NE flow direction (Fig. 10a) suggests the deposits coming from SW (Fig. 1B) of the study section. Further support for this SW direction is given by the first-order lithofacies distribution of the Middle Triassic basin wide, which presents a proximal to distal transition from south to north as demonstrated by slump or slide deposits (like clay-boulders) on the southern margin of the basin and generally fine lithologies in the central-northern basin (GXGMR, 1985; GZGMR, 1987; Huang, 1998; Lepvrier et al., 2011).

The SW subaqueous source area may be adjacent to a predominant subaerial erosional provenance (e.g., the northern Vietnam terrane) in the simplest case where no intervening long river systems exist between the provenance and depositional basin. The inference is supported by several other aspects of the siliciclastic rocks that argue for a single, nearby, recycled orogenic source located south of the Nanpanjiang Basin. For the studied section, this suite of siliciclastic rocks shows a relatively constant composition of magnetic minerals as indicated by essentially normal distributed bulk susceptibility and NRM values (Fig. 5a and b), which in turn may mean a single erosional source involved. This is also reflected on monotonic detrital framework in most part of the basin that is dominated by quartz and lithic fragments with extremely low feldspar contents (Gou, 1985; GXGMR, 1985; Ceng et al., 1995; Cai, 2009). The angular appearance of framework grains indicates short distance transport and hence a nearby erosional source; while the framework mode is characteristic of recycled orogens covered by sedimentary sequences (Dickinson and Suczek, 1979; Dickinson et al., 1983), thus excluding the Yunkai massif or the Jiangnan oldland (Fig. 1B) as major contributor given their dominant basement metamorphic complexes and igneous rocks. Like the case in the Songpan-Ganzi basin in northern Tibet (Gu, 1994; Bruguier et al., 1997), the general origin from sedimentary covers does not exclude minor constituents from ocean crust such that detrital zircons (Wood et al., 2012; Yang et al., 2012) extracted from this suite of clastic rocks have an age structure typified by Paleo-Tethys sutured between the South China and Indochina blocks (Cai and Zhang, 2009). Therefore, these lines of evidence together point to the northern Vietnam terrane, a more or less gravity-flow orthogonal orogenic provenance, witnessing an episode of mountain-building movement.

6.3. Orogeny and deposition

Collisional orogenesis is traditionally treated as one of main controlling factors in driving lithospheric flexural bending. In such a setting, the uplifting bulges will be eroded to shed detritus into adjacent subsiding basins (e.g., Dickinson and Suczek, 1979; Dickinson et al., 1983; Bourgeois et al., 2007). Accordingly, tracing the source regions of clastic deposits in the sedimentary basin can help to pinpoint the location of orogens as well as to constrain the associated tectonic processes. By such rationale, the SE or E provenance proposed in previous studies (Xia et al., 1993; Ceng et al., 1995; Gou, 1985) has been correlated to a southeastern South China orogen formed during, for example, the NW-directed subduction of the Paleo-Pacific plate beneath the eastern Asian margin in the early Mesozoic (Carter and Clift, 2008). However, such early NW–SE convergence is at odds with the WNW–ESE to NW–SE trending of folding and thrusting in southern South China that have been dated at ages of the Triassic through $^{40}Ar/^{39}Ar$ analysis on muscovites (250–255 Ma) and biotites (208–230 Ma) extracted from mylonites (cf., Lin et al., 2008 and references therein). Besides, this SE provenance is not supported by the paleocurrent evidence presented here.

The SW provenance inferred from paleocurrent analysis in the study tends to suggest a southwestern orogen (Fig. 1B). The orogeny is confirmed by the WNW–ESE to NW–SE trending of fold-and-thrust structures with Triassic ages (236–198 Ma) present on the southern margin of the basin and in the adjacent northern Vietnam terrane (Wu et al., 1999; Roger et al., 2000; Maluski et al., 2001; Wu, 2002; Lepvrier et al., 2011). These structures present an overall NNE to NE vergence consistent with that inferred from soft sediment deformation in the Middle Triassic sequence within the basin (Lv et al., 2003). Further support for the orogeny comes from the Early to Middle Triassic paleogeographic reconstruction for isolated carbonate platforms within the basin that concluded a lithospheric flextural bending beginning from the southern margin of the basin (Lehrmann et al., 2007). According to the plate framework in the Late Paleozoic and Early Mesozoic times (e.g., Metcalfe, 2002; Li et al., 2004), the only plate available immediately south of the South China block was the Indochina block. Therefore, the origin of siliciclastic rocks within the Nanpanjiang Basin should be related to the Indosinian (P_2-T) convergent orogeny between the

Indochina and South China blocks (Cai et al., 2008; Cai and Zhang, 2009).

However, there is a divide in literature as to the mode of the Indosinian orogenesis. Some authors consider it a tectonic reactivation between the already amalgamated South China and Indochina blocks caused by oblique accretion of the Sibumasu block onto the Indochina block from the late Permian through Triassic (Carter and Clift, 2008; Carter et al., 2001). The amalgamation between the two blocks occurred along the Song Ma suture much earlier in the meso-Paleozoic time (e.g., Metcalfe, 2002). In contrast, most previous studies have supported the interpretation of Indosinian collision between them, although with various proposals as to the location of suturing (Fig. 11a), such as the Song Ma zone (Lepvrier et al., 2004), the Song Ma-Song Hong region (Findlay and Phan, 1997), the Red River fault zone (Lepvrier et al., 2011; Tapponnier et al., 1990), or the Dian-Gui-Qiong zone (Zhong et al., 1998; Wu et al., 1999; Cai and Zhang, 2009). These interpretations are further complicated by the complex tectonics in central-northern Vietnam, especially those related to the large-scale wrench movement along the Red River fault zone since the Oligocene (e.g., Tapponnier et al., 1990; Findlay and Phan, 1997).

Here, we prefer a separate configuration of the Indochina and South China blocks prior to the Triassic given the sporadic mafic or ultramafic rocks with island, island arc, and mid-ridge characters and late Paleozoic ages along the Dian-Gui-Qiong suture (e.g., Wu et al., 1999; Cai and Zhang, 2009; Yang et al., 2012). The Qiong part of this suture has been displaced into its present location on Hainan Island by a dextral strike-slip system, called the Guangning-Bobai fault zone (Fig. 11a), in the Early Jurassic (Cai, 2013). Thus, the Middle Triassic siliciclastic deposits in the Nanpanjiang Basin were originated from the Indosinian collisional orogeny between the Indochina and South China blocks.

6.4. Paleogeographic reconstruction

A tropic environment has prevailed in the Nanpanjiang region during the deposition of the Middle Triassic siliciclastic rocks. The paleomagnetic results in this study indicate a low latitude, ca. N8.6°, for the studied cross-section, slight higher than that (N3.1°) calculated using the mean Early-Middle Triassic paleomagnetic pole for the South China block (Gilder et al., 2008). Such a tropic environment is in favor of rapid erosion and deposition of clastic detritus with a low textural but a relatively high compositional maturity (e.g., van Hattum et al., 2006) and also can explain the coeval carbonate deposition on the isolated carbonate platforms within (Lehrmann et al., 2005, 2007) and on the margins of the Nanpanjiang Basin (Enos et al., 1998).

A general N–S convergent orogeny with an overall N-directed sediment-gravity flow regime has occurred during the deposition of the Middle Triassic siliciclastic rocks. Our paleomagnetic results indicate that the studied Middle Triassic cross-section shows no remarkable local or regional rotation relative to the South China block, a conclusion also proposed by previous studies in the Nanpanjiang Basin (Gilder et al., 1995, 2008). The ca. N45°E magnetic declination indicate the studied region together the South China block as a whole has rotated clockwise since the Middle Triassic. Restoring this block rotation will render the overall NE–SW convergent Indosinian orogeny and paleocurrent direction to be nearly N–S.

Therefore, we propose the following scenarios. The secular stagnation of the South China block across the equatorial zone since the Late Paleozoic began to change in the Late Permian (e.g., Metcalfe, 2002; Li et al., 2004), accompanying an episode of continent-building movement that finally formed the southeastern part of the Eurasia continent in the Late Triassic (e.g., Goldfarb et al., 2001; Golonka, 2007). Several continent blocks began to drift northwards and were amalgamated as manifested by the convergence between the Indochina and South China blocks since the Late Permian (Cai and Zhang, 2009). The continuing N–S collision of the two blocks along the Dian-Gui suture in the Middle Triassic resulted in subsidence of the Nanpanjiang region to form a foreland or foredeep basin to receive voluminous debris shed from its uplifting southern margin (Fig. 11b). Accompanying the northward advance of the orogenic front, the basin

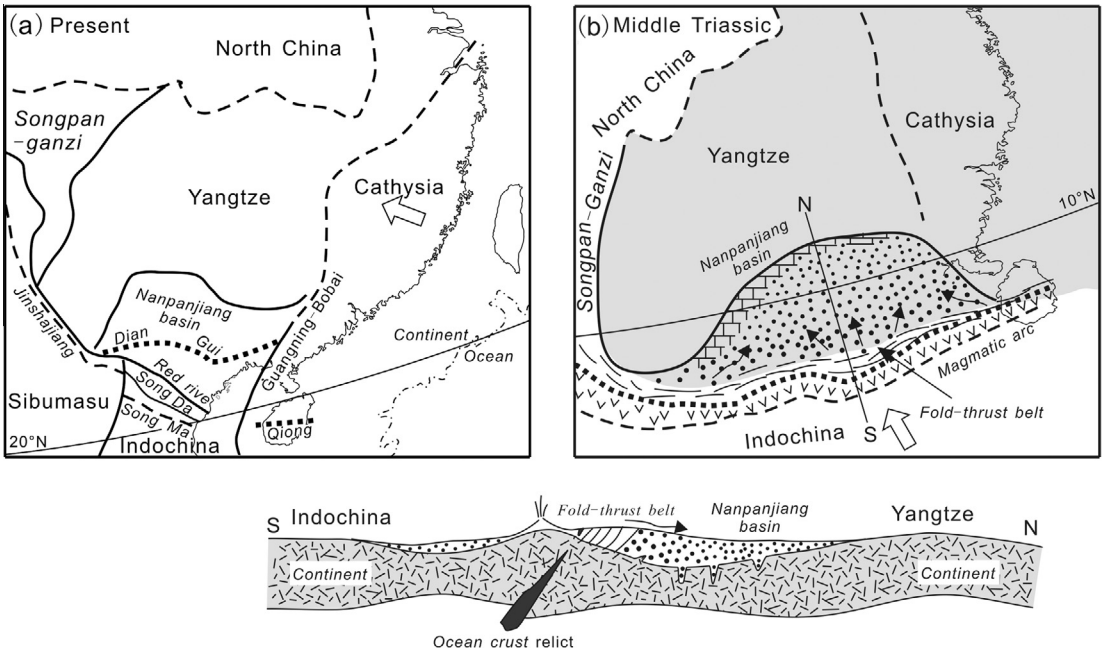


Fig. 11. Cartoons showing plate tectonics of the South China plate. (a) Present configuration of crustal blocks in SE Asia after Cai and Zhang (2009). (b) Tectonic reconstruction of the southern South China block in the Middle Triassic. Note the ca. 45° anticlockwise rotation. Plane (top-right) and profile (bottom) views showing nearly N–S convergence between the Indochina and South China blocks that caused the subsiding Nanpanjiang foreland basin to receive detritus eroded from its uplifting southern margin. The profile redrawn after Dickinson and Suczek (1979) and Bourgeois et al. (2007).

region was uplifted by the Late Triassic, receiving shallow-sea to braided-stream deposits from south to north.

7. Conclusion

The Middle Triassic siliciclastic rocks in the Nanpanjiang Basin in southern South China have been investigated in terms of paleoflow and provenance. These clastic rocks have an angular detrital framework, comprising mainly quartz and lithic fragments with extremely low feldspar contents, diagnostic of a nearby, recycled orogenic origin. The relatively monotonic magnetic mineral composition as demonstrated by the normal distribution of bulk susceptibility and natural remanent magnetization may indicate a single provenance involved in deposition. AMS analysis on clastic rocks from the Tiangyang-Bama cross-section reveals paleocurrent directions from NNW to ENE with an overall NE orientation. Demagnetization analysis acquires a mean paleomagnetic direction consistent with that expected from the South China block, implying a ca. 45° clockwise rotation for the study region as well as the South China block as a whole. The paleocurrent direction is essentially in the N–S direction when the paleogeography is reconstructed, thus pointing to a southern orogenic source region. Given coeval plate framework, this study proposes Indosinian collision between the Indochina and South China blocks responsible for the origin of the siliciclastic rocks in the Nanpanjiang Basin.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jseaes.2013.10.033.

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